

The thorax is the upper part of the trunk. It consists of an external musculoskeletal cage, the thoracic wall, and an internal cavity that contains the heart, lungs, oesophagus, trachea and principal bronchi, thymus, vagus and phrenic nerves, right and left sympathetic trunks, thoracic duct, lymph nodes, and major systemic and pulmonary blood vessels. Inferiorly, the thorax is separated from the abdominal cavity by the diaphragm; superiorly, it communicates with the neck and the upper limbs. The thoracic wall offers protection to some of the abdominal viscera: the greater part of the liver lies under the right dome of the diaphragm; the stomach and spleen lie under the left dome of the diaphragm; and the posterior aspects of the superior poles of the kidneys lie on the diaphragm and are anterior to the twelfth rib on the right, and to the eleventh and twelfth ribs on the left (Fig. 51.1).

Variations in thoracic dimensions and proportions are partly individual and also linked to age, sex and race. At birth, the transverse diameter is relatively less than it is in the adult, but adult proportions develop as walking begins. Thoracic capacity is less in females than it is in males, both absolutely and proportionately: the female sternum is shorter, the thoracic inlet more oblique, and the suprasternal notch is level with the third thoracic vertebra, whereas it is level with the second thoracic vertebra in males. In all individuals, the size of the thoracic cavity changes continuously according to the movements of the ribs and diaphragm during respiration (Ch. 55) and the degree of abdominal visceral distension.

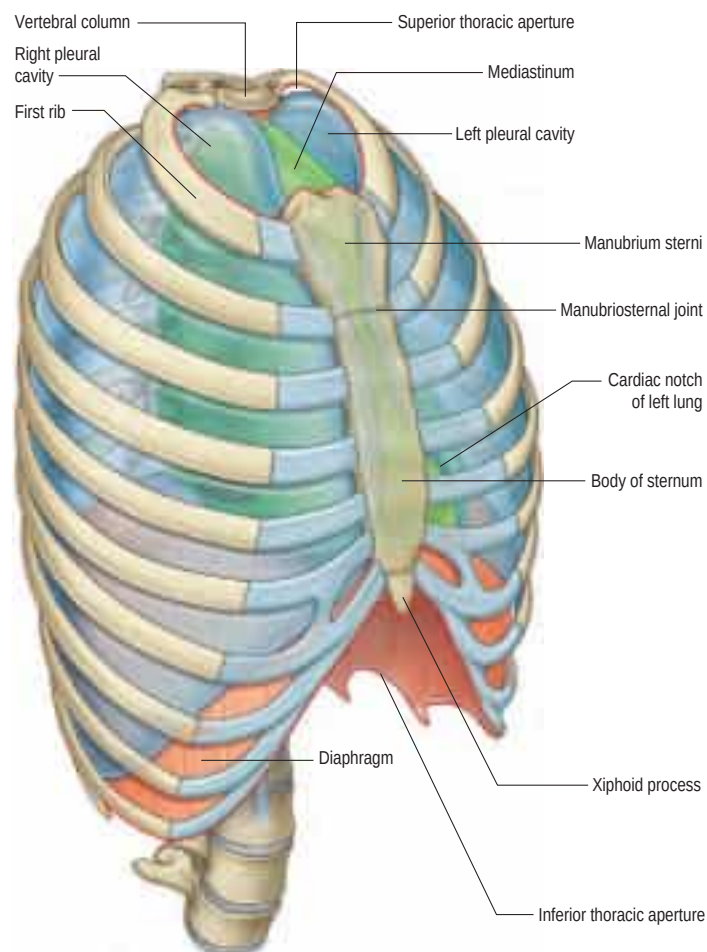


Fig. 51.1 The thoracic wall and cavity. (With permission from Drake RL, Vogl AW, Mitchell A (eds), *Gray's Anatomy for Students*, 2nd ed, Elsevier, Churchill Livingstone. Copyright 2010.)

MUSCULOSKELETAL FRAMEWORK

BONES AND JOINTS

The thoracic skeleton consists of twelve thoracic vertebrae and their intervening intervertebral discs (midline, posterior), twelve pairs of ribs and their costal cartilages (predominantly lateral), and the sternum (midline, anterior). When articulated, they form an irregularly shaped osteocartilaginous cylinder, reniform in horizontal section, which is narrow above, broad below, flattened anteroposteriorly and longer behind (see Fig. 53.8). Laterally, the thoracic cage is convex and is formed by the ribs; anteriorly, it is slightly convex and is formed by the sternum and the distal parts of the ribs and their costal cartilages. The first seven pairs of ribs are connected to the sternum by costal cartilages, the costal cartilages of the eighth to tenth ribs usually join the superjacent cartilage, and the eleventh and twelfth ribs are free (floating) at their anterior ends. The posterolateral curvature of the ribs, from their vertebral ends to their angles, produces a deep internal groove, the paravertebral gutter, on either side of the vertebral column. The ribs and costal cartilages are separated by intercostal spaces, which are deeper anteriorly and between the upper ribs. Each space is occupied by three layers of flat muscles and their aponeuroses, neurovascular bundles and lymphatic channels.

The narrow thoracic inlet (superior thoracic aperture) typically measures 5 cm anteroposteriorly and 10 cm transversely. It is bounded by the first thoracic vertebral body posteriorly, the medial border of the first ribs on each side, and the superior border of the manubrium sterni anteriorly. It slopes down and forwards, so that the apex of the lung extends upwards into the neck behind the anterior end of the first rib; structures that pass between the thorax and the upper limb therefore pass over the first rib and the apices of the lungs and the apical pleurae.

The broad thoracic outlet (inferior thoracic aperture) is limited posteriorly by the twelfth thoracic vertebral body, posterolaterally by the twelfth rib and the distal end of the eleventh rib, anterolaterally by the distal cartilaginous ends of the seventh to tenth ribs (which unite and ascend to form the costal margin), and anteriorly by the xiphoid process. It is wider in the transverse plane than in the sagittal plane and slopes obliquely inferiorly and posteriorly, so that the thoracic cavity is deeper posteriorly than anteriorly.

The horizontal sternal plane is traditionally reported to pass through the intervertebral disc between the fourth and fifth thoracic vertebrae posteriorly and the manubriosternal joint (sternal angle or angle of Louis) at the level of the second costal cartilage anteriorly. Recent data place the sternal angle between the fourth thoracic vertebra and the upper half of the fifth thoracic vertebra in the majority of adults (Mirjalili et al 2012a). The sternal plane separates the superior mediastinum from the inferior mediastinum and marks the positions of the superior limits of the pericardium; the bifurcation of the pulmonary trunk; the origin of the arch of the aorta; the level at which the trachea bifurcates into right and left principal bronchi; and the site where the superior vena cava penetrates the pericardium to enter the right atrium.

MUSCLES

Intrinsic and extrinsic muscles

The intrinsic muscles of the chest wall are the intercostal muscles, subcostalis, transversus thoracis, levatores costarum and serratus posterior superior and inferior.

The uncommon variants, sternalis and rectus thoracis bifurcalis, are important clinically as they may mimic a focal density in medial breast craniocaudal mammograms and be encountered during reconstructive surgery of the breast and chest wall (p. 942).

The intercostal muscles occupy each of the intercostal spaces and are named according to their surface relations, i.e. external, internal and innermost. All except levatores costarum are innervated by the adjacent intercostal nerves derived from the ventral rami of the thoracic spinal nerves; levatores costarum are innervated by the dorsal rami of the thoracic spinal nerves. The intrinsic muscles can elevate or depress the rib and are particularly active in forced respiration; their primary action is believed to be to stiffen the chest wall, preventing paradoxical movement during inspiration.

The skeletal framework of the thoracic wall provides extensive attachment sites for muscles associated functionally with the neck, abdomen, back and upper limbs. Some of them (scalenes, infrahyoid strap muscles, sternocleidomastoid, serratus anterior, pectoralis major and minor, external and internal obliques, and rectus abdominis) function as accessory muscles of respiration and are usually active only during forced respiration; scalenus medius is active in quiet inspiration. Scalenus anterior, medius and posterior are described in [Chapter 29](#). Trapezius, latissimus dorsi, rhomboid major, rhomboid minor, levator scapulae, pectoralis major and minor, subclavius and serratus anterior are described in [Chapter 48](#). Rectus abdominis, external oblique and internal oblique are described in [Chapter 61](#).

Diaphragm

The diaphragm, a curved musculotendinous sheet attached to the circumference of the thoracic outlet and to the upper lumbar vertebrae, forms the floor of the thoracic cavity, separating it from the abdominal cavity ([Ch. 55](#)). The diaphragm is relatively flat centrally and domed peripherally, rising higher on the right side than on the left, an asymmetry that reflects the relative densities of the underlying liver and gastric fundus, respectively. From its most superior point on each side, the diaphragm slopes inferiorly to its costovertebral attachments; this slope is most marked posteriorly, where the space between the diaphragm and the posterior wall of the thorax is very narrow. The domes are higher in the supine position. During quiet respiration in the erect position, diaphragmatic excursion is about 2 cm in both sexes, increasing to around 7 cm during deep breathing ([Boussuges et al 2009](#)); values are greater in men and possibly increase further in the supine position ([Takazakura et al 2004](#)).

Diaphragmatic openings

The inferior vena cava passes through an opening in the central tendon of the diaphragm to enter the right side of the mediastinum at the level of the eighth thoracic vertebra; the oesophagus passes through the muscular part of the diaphragm and enters the abdomen just to the left of the midline at the level of the tenth thoracic vertebra; and the descending thoracic aorta passes posterior to the diaphragm in the midline at the level of the twelfth thoracic vertebra (see [Fig. 55.1](#)). Other structures that pass between the thorax and abdomen include the right and left vagi, phrenic and subcostal nerves, thoracic duct, sympathetic trunks, thoracic splanchnic nerves, right and left superior epigastric arteries, and the hemiazygos and accessory hemiazygos veins.

According to a CT study, the inferior vena cava and the oesophagus traverse the diaphragm at the upper and lower borders, respectively, of the eleventh thoracic vertebra; these levels are lower than those based on cadaveric/radiographic studies (eighth and tenth thoracic vertebrae, respectively) ([Mirjalili 2012a](#)).

THORACIC CAVITY

PLEURAL CAVITIES

The right and left pleural cavities are separate compartments on either side of the mediastinum. Each encloses a lung and its associated bronchial tree, and vessels, nerves and lymphatics ([Ch. 54](#)). The walls are formed by a serous membrane, the pleura, arranged as a closed sac. The outer layer of the sac, the parietal pleura, lines the corresponding half of the thoracic wall and covers much of the diaphragm and structures occupying the middle region of the thorax. The inner or visceral layer is more delicate and adheres closely to the pulmonary surface, following the interlobar fissures. The two layers are continuous with each other around the hilar structures, and remain in close, though sliding, contact at all phases of respiration. The potential space between them is the pleural cavity, which is maintained at a negative pressure by the inward elastic recoil of the lung and the outward pull of the chest wall. The lungs do not fill this space in quiet respiration, but move into

recesses such as the costodiaphragmatic recess, which separates the costal and diaphragmatic pleura, in deep breathing.

The left pleural cavity is the smaller of the two pleural cavities because the heart extends further to the left.

MEDIASTINUM

The mediastinum lies between the right and left pleural sacs in and near the median sagittal plane of the chest ([Ch. 56](#)). It extends from the sternum anteriorly to the vertebral column posteriorly. A horizontal plane passing through the manubriosternal joint, and the intervertebral disc between the fourth and fifth thoracic vertebrae separates the mediastinum into superior and inferior portions.

Superior mediastinum

The superior mediastinum lies between the manubrium sterni anteriorly and the upper thoracic vertebrae posteriorly, and is limited laterally by the pleurae. Its inferior boundary is a slightly oblique plane that passes backwards from the manubriosternal joint to the lower part of the body of the fourth thoracic vertebra. It contains the lower ends of sternohyoid, sternothyroid and longus colli on each side; the aortic arch, brachiocephalic trunk and thoracic portions of the left common carotid and left subclavian arteries; the brachiocephalic veins and upper half of the superior vena cava; the left highest intercostal vein; the vagus, cardiac, phrenic and left recurrent laryngeal nerves; and the trachea, oesophagus, thoracic duct, thymic remnants and lymph nodes.

Inferior mediastinum

The inferior mediastinum is divided into three parts: anterior, in front of the pericardium; middle, containing the pericardium and its contents ([Ch. 57](#)); and posterior, behind the pericardium.

Anterior mediastinum The anterior mediastinum exists only on the left side where the left pleura diverges from the mid-sternal line. It is bounded anteriorly by the sternum, laterally by the pleurae, and posteriorly by the pericardium. It is narrow above but widens a little below. Its anterior wall is formed by the left transversus thoracis and the fifth to seventh left costal cartilages. It contains a quantity of loose areolar tissue, lymphatic vessels that ascend from the convex surface of the liver, two or three anterior mediastinal lymph nodes, and the small mediastinal branches of the internal thoracic artery.

Middle mediastinum The middle mediastinum is the broadest part of the interpleural space; it contains the heart enclosed in the pericardium; the ascending aorta; the lower half of the superior vena cava with the azygos vein opening into it; the bifurcation of the trachea and the right and left principal bronchi; the pulmonary artery dividing into its two branches; the right and left pulmonary veins and phrenic nerves; and some bronchial lymph nodes.

Posterior mediastinum The posterior mediastinum is an irregular triangular space, bounded anteriorly by the pericardium above and the posterior surface of the diaphragm below, posteriorly by the vertebral column from the lower border of the fourth to the twelfth thoracic vertebrae, and laterally by the mediastinal pleura. It contains the thoracic part of the descending aorta, the azygos and hemiazygos veins, the vagus and splanchnic nerves, oesophagus, thoracic duct and lymph nodes.

VASCULAR SUPPLY AND LYMPHATIC DRAINAGE

ARTERIES

The skin of the thorax is supplied by direct cutaneous vessels and musculocutaneous perforators, which reach the skin primarily via the intercostal muscles, pectoralis major, latissimus dorsi and trapezius. Branches from the thoraco-acromial axis, lateral thoracic, internal thoracic, anterior and posterior intercostal, thoracodorsal, transverse cervical, dorsal scapular and circumflex scapular arteries are the major contributing vessels (see [Figs 43.4, 53.1–53.2](#)).

Muscles of the thoracic wall receive their blood supply from the internal thoracic artery (either directly or via the musculophrenic artery), the superior intercostal artery (from the costocervical trunk), superior thoracic artery (from the axillary artery), descending thoracic

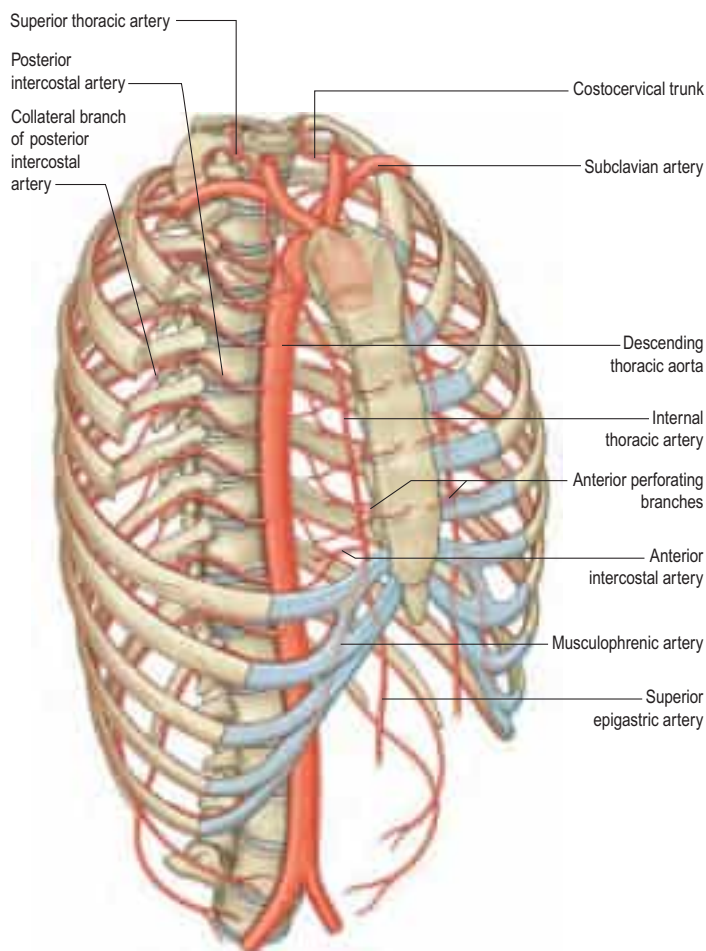


Fig. 51.2 Arteries of the thoracic wall and thoracic aorta. (With permission from Drake RL, Vogl AW, Mitchell A, Tibbitts R, Richardson P (eds), Gray's Atlas of Anatomy, Elsevier, Churchill Livingstone. Copyright 2008.)

aorta, and the subcostal artery (Fig. 51.2). Additional contributions come from vessels that supply the proximal muscles of the upper limb, namely: suprascapular, superficial cervical, thoraco-acromial, lateral thoracic and subscapular arteries.

The detailed arterial supply of the thoracic viscera is described with the viscera.

VEINS

The intercostal veins accompany the similarly named arteries in the intercostal spaces (Fig. 51.3). The small anterior intercostal veins are tributaries of the internal thoracic and musculophrenic veins; the internal thoracic veins drain into the appropriate brachiocephalic vein. The posterior intercostal veins drain backwards, and most drain directly or indirectly into the azygos vein on the right and the hemiazygos or hemiazygos veins on the left. The azygos veins exhibit great variation in their origin, course, tributaries, anastomoses and termination.

LYMPHATIC DRAINAGE

Superficial lymphatic vessels of the thoracic wall ramify subcutaneously and converge on the axillary nodes (see Fig. 53.22). Lymph vessels from deeper tissues of the thoracic walls drain mainly to the parasternal, intercostal and diaphragmatic lymphatic nodes.

INNERVATION

THORACIC SPINAL NERVES

There are twelve pairs of thoracic spinal nerves. The ventral rami, unlike their cervical and lumbar counterparts, have retained a largely segmental distribution to the body wall (see Figs 16.10, 53.3). The upper eleven lie between the ribs (intercostal nerves), and the twelfth lies below the last rib (subcostal nerve). Each is connected with the adjoining ganglion

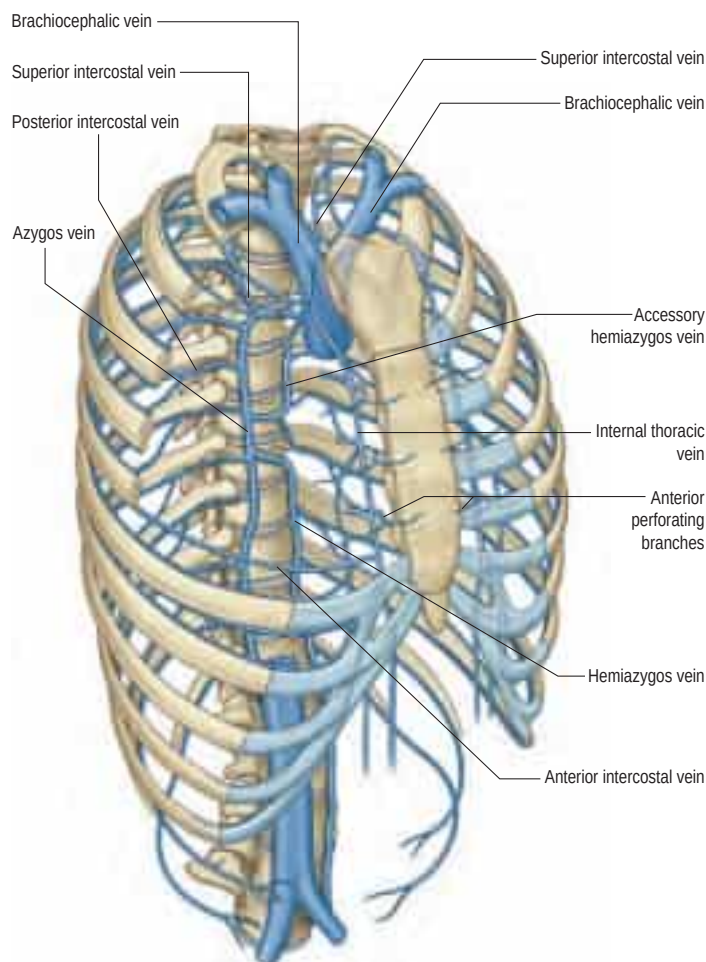


Fig. 51.3 Veins of the thoracic wall and azygos system. (With permission from Drake RL, Vogl AW, Mitchell A, Tibbitts R, Richardson P (eds), Gray's Atlas of Anatomy, Elsevier, Churchill Livingstone. Copyright 2008.)

of the sympathetic trunk by pre- and postganglionic branches (white and grey rami communicantes, respectively). Intercostal nerves are distributed primarily to the thoracic and abdominal walls. The greater part of the first thoracic ventral ramus passes into the brachial plexus, together with a variable proportion of the second. The next four ventral supply only the thoracic wall, and the lower five supply both thoracic and abdominal walls. The subcostal nerve is distributed to the abdominal wall and the gluteal skin. Communicating branches link the intercostal nerves posteriorly in the intercostal spaces, and the lower five nerves communicate freely in the abdominal wall. Thoracic dorsal rami divide into medial and lateral branches that supply the intrinsic muscles of the back and the overlying postvertebral skin on a segmental basis (see Fig. 43.3).

The supraclavicular branches of C4 from the cervical plexus innervate the skin over the clavicle and may communicate with the anterior cutaneous branches of the second thoracic ventral ramus (p. 443).

AUTONOMIC INNERVATION

The autonomic nervous system in the thorax consists of right and left sympathetic trunks and vagus nerves, and the cardiac, oesophageal and pulmonary plexuses.

Sympathetic trunks

The ganglionated sympathetic trunks lie anterior to the heads of the ribs; the ganglia are arranged segmentally (see Fig. 56.6). Preganglionic sympathetic axons originate from neurones in the lateral grey column of the spinal cord from T1 to L2, and leave the spinal cord with the corresponding ventral roots as white rami communicantes. Their targets vary. Some enter the sympathetic trunk, where they either synapse in their segmental ganglion, or ascend or descend in the trunk to synapse in cervical or lumbar ganglia. Many of the preganglionic axons that

originate in the lower thoracic spinal segments (T5–12) do not synapse locally but enter the abdominal cavity as the thoracic splanchnic nerves; they synapse either in prevertebral ganglia, especially the coeliac ganglion (Ch. 59), or around the medullary chromaffin cells of the suprarenal gland (Ch. 71). Axons destined for the cervical ganglia are derived from preganglionic neurones in T1.

Vagus nerves in the thorax

Preganglionic parasympathetic fibres arise from neuronal cell bodies in the dorsal motor nucleus of the vagus in the medulla. Those destined for the thoracic viscera travel in the pulmonary, cardiac and oesophageal branches of the vagus and synapse in minute ganglia in the visceral walls. Axons travelling in cardiac branches join the cardiac plexuses and synapse in ganglia distributed over both atria; when stimulated, they slow the cardiac cycle (Ch. 57). Pulmonary branches contain axons that relay in ganglia of the pulmonary plexuses; they are motor (bronchoconstrictor) to the circular non-striated muscle fibres of the bronchi and bronchioles, and secretomotor to the mucous glands of the respiratory epithelium.

Autonomic plexuses in the thorax

Cardiac plexus The cardiac plexus has a superficial component inferior to the aortic arch, lying between it and the pulmonary trunk, and a deep part between the aortic arch and tracheal bifurcation (Ch. 57). The superficial part is formed by the cardiac branch of the left superior cervical sympathetic ganglion and the lower of the two cervical cardiac branches of the left vagus. The deep part is formed by the cardiac branches of the cervical and upper thoracic sympathetic ganglia and of the vagus and recurrent laryngeal nerves. The only cardiac nerves that do not join it are those that join the superficial part of the plexus. Branches from the cardiac plexuses also form the left and right coronary and atrial plexuses.

Pulmonary plexus The pulmonary plexuses are anterior and posterior to the other structures at the hila of the lungs (Ch. 54). They are formed by cardiac branches from the second to fifth (or sixth) thoracic sympathetic ganglia and from the vagus and cervical sympathetic cardiac nerves. The left plexus also receives branches from the left recurrent laryngeal nerve.

Oesophageal plexus The oesophageal plexus surrounds the oesophagus below the level of the lung roots (Ch. 56). Vagal fibres either pass through the plexus or are given off directly by the vagus in the thorax. All fibres relay in the oesophageal wall and are motor to the smooth muscle in the lower oesophagus and secretomotor to mucous glands in the oesophageal mucosa. Vasomotor sympathetic fibres arise from the upper six thoracic spinal cord segments. Those from the upper segments synapse in cervical ganglia; postganglionic axons innervate the vessels of the cervical and upper thoracic oesophagus. Fibres from the lower segments pass directly to the oesophageal plexus or to the coeliac ganglion, where they synapse; postganglionic axons innervate the vessels of the distal oesophagus.

SURFACE ANATOMY

In clinical practice, an evidence-based knowledge of thoracic surface anatomy underpins practical procedures such as placing drains or catheters and siting surgical incisions. The accuracy of some older descriptions of clinically important surface landmarks, based on cadaveric or radiographic studies, has been questioned recently (Hale et al 2010). Cross-sectional computed tomography (CT) imaging, undertaken with subjects supine, at end tidal inspiration, and with arms abducted, has emphasized a greater inter-individual variation in some thoracic surface anatomy landmarks than appears in the current anatomical literature (Mirjalili et al 2012a).

BREAST

In females, the base of the breast is relatively constant in position; it extends vertically from the second or third to the sixth rib, and from the lateral sternal border medially almost to the mid-axillary line laterally. The tail of the breast extends towards the axilla along the inferolateral border of pectoralis major (Ch. 53). In adult females, the site of

the nipple is dependent on the size and shape of the breasts. In adult males, the nipple is usually sited either in the fourth intercostal space or over the fifth rib in the mid-clavicular line, approximately 20 cm from the sternal notch and mid-clavicular point (Beckenstein et al 1996). The pigmented areola surrounding the nipple varies in size.

SKELETAL LANDMARKS

Anteriorly, the clavicle, sternoclavicular and acromioclavicular joints are both palpable and visible in all but obese individuals (Fig. 51.4). The sternum may be felt throughout its length in the midline, although laterally it may be covered by pectoralis major. The jugular (suprasternal) notch is palpable in the midline at the superior sternal border and typically lies at the level of the junction between the second and third thoracic vertebrae. The cartilaginous rings of the trachea are palpable immediately superior to the notch, using a posteriorly directed finger. The manubriosternal angle is more pronounced in the male than in the female, and is palpable at the junction of the manubrium with the sternal body. It serves as a useful landmark because it indicates the level of the sternal plane and the medial ends of the second costal cartilages, and so offers an accurate point at which to start counting ribs. In the majority of adults, the sternal angle is located between the fourth thoracic vertebra and the upper half of the fifth thoracic vertebra, with a range from the lower half of the second thoracic vertebra to the lower half of the sixth thoracic vertebra, a linear distance of about 8 cm; the plane lies at a slightly higher vertebral level in women. The xiphisternal joint and xiphoid process are palpable at the inferior end of the sternum; the joint usually lies at the level of the ninth thoracic vertebra. The costal margin is palpable passing inferolaterally from the xiphisternum. Posteriorly, the free ends of the eleventh and twelfth ribs may be

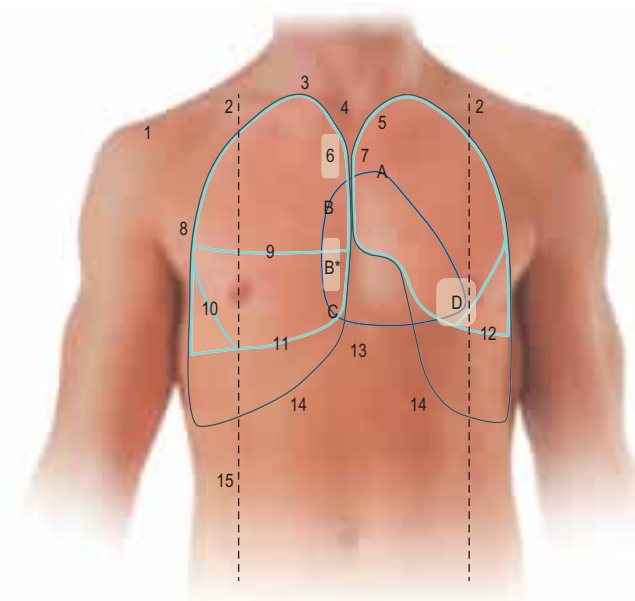


Fig. 51.4 Surface anatomy of the anterior thorax. Key: 1, right acromioclavicular joint; 2, clavicle and mid-clavicular line (dashed black); 3, apex of right lung, located posterior to the medial third of the clavicle; 4, sternal notch of manubrium sterni: the trachea may be located here by posterior palpation; 5, sternoclavicular joint: marks the junction of the internal jugular and subclavian veins to form the brachiocephalic vein; 6, zone of formation of the superior vena cava: from first intercostal space to second costal cartilage level (78% of subjects, white zone); 7, sternal angle: marks the level of the sternal plane and the second costal cartilage; 8, pectoralis major and the anterior axillary fold; 9, horizontal fissure; 10, right oblique fissure; 11, lower anterior border of the right lung: typically, either the sixth intercostal space or the seventh rib in the mid-clavicular line; 12, lower anterior border of the left lung: typically, either at the fifth rib or fifth intercostal space in the mid-clavicular line; 13, xiphisternum; 14, costal margin; 15, tenth costal cartilage, forming the lower part of the costal margin. A–D and blue line: conventional surface markings of the heart. A overlies the left second costal cartilage; B overlies the right third costal cartilage; B*, zone of the superior vena cava meeting the right atrium from the fourth to fifth costal cartilage in 71% of subjects at end tidal inspiration (from CT data); C, right sixth costal cartilage; D, zone of location of the cardiac apex: almost always from the level of the fifth to the sixth rib, and a mean of 8.7 ± 1 cm from the midline.

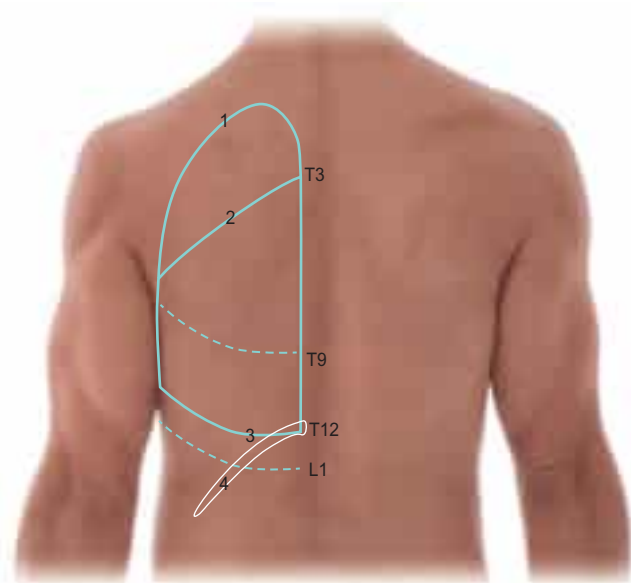


Fig. 51.5 Surface anatomy of the posterior thorax. The levels of the spinous processes of the third, ninth and twelfth thoracic vertebrae and the first lumbar vertebra are indicated in the midline. Key: 1, border of the left lung; 2, oblique fissure: passes anteroinferiorly from the spinous process of the third thoracic vertebra to cross the fifth rib in the mid-axillary line. The upper lobe sits superiorly and the lower lobe inferiorly; 3, lower border of the lung: this is often located at level of the twelfth thoracic vertebra but may be lower, at the level of the first lumbar vertebra, adjacent to the vertebral column at end tidal inspiration. Dashed blue lines indicate the range of levels for the lower border of the lung (ninth thoracic vertebra to first lumbar vertebra); 4, twelfth rib: can be traced superomedially to aid identification of the spinous process of the twelfth thoracic vertebra.

palpable. Tracing the twelfth rib superomedially aids identification of the spinous process of the twelfth thoracic vertebra (Fig. 51.5). In thin individuals, it is possible to palpate all the ribs anteriorly, from the first to the costal margin. Posteriorly, the spinous processes of the thoracic vertebrae are palpable; the spinous process of the first thoracic vertebra sits below that of the seventh cervical vertebra (vertebra prominens) and is often more prominent. The angles of the ribs are palpable several centimetres lateral to the spinous processes of the vertebrae.

The surface anatomy of the back is described in Chapter 43.

MUSCULOTENDINOUS LANDMARKS

In a lean, muscular subject, pectoralis major, the slips of serratus anterior, latissimus dorsi, trapezius, external oblique and rectus abdominis are all easily visible. In females, the breasts may significantly obscure the anterior thoracic wall.

INTRATHORACIC VISCERA

The sternal plane represents the lower border of the superior mediastinum. At this point, the right and left pleurae are in contact with each other; it is therefore a useful starting point when delineating the surface markings of the parietal pleura (see Fig. 51.7). The sternal plane is conventionally described as lying over the tracheal bifurcation, the concavity of the aortic arch and the point where the azygos vein enters the superior vena cava. CT analysis places these three major surface landmarks typically a little lower, at the level of the fifth or sixth thoracic vertebra rather than in the plane of the sternal angle (between the fourth and fifth thoracic vertebrae) (Mirjalili et al 2012a) (Fig. 51.6; Table 51.1).

Heart

The cardiac surface projections described below apply to an average adult. They may be modified by age, sex, stature, ventilation, the position of the diaphragm and posture (Macklin 1925). The projection of the cardiac borders on to the anterior thoracic wall forms a trapezoid. The area of cardiac dullness may be mapped out via percussion.

The upper border slopes gently from the second left costal cartilage to the third right costal cartilage (see Fig 51.4). CT analysis places the

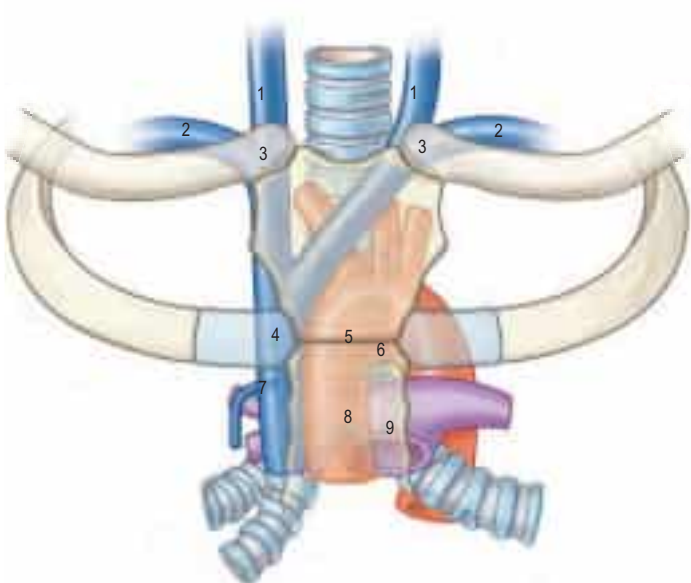


Fig. 51.6 Surface anatomy of the great vessels and tracheobronchial tree relative to bony landmarks. Key: 1, internal jugular vein; 2, subclavian vein; 3, formation of the brachiocephalic vein posterior to the sternoclavicular joints; 4, formation of the superior vena cava, posterior to the right second costal cartilage or first intercostal space; 5, manubriosternal joint, 6, concavity of the aortic arch, typically sitting inferior to the sternal plane, level with the upper half of the fifth thoracic vertebra; 7, azygos vein entering the superior vena cava: typically, it sits inferior to the sternal plane, level with the lower half of the fifth thoracic vertebra; 8, tracheal bifurcation: typically, it sits inferior to the sternal plane, level with the upper half of the sixth thoracic vertebra; 9, bifurcation of the pulmonary trunk, level with the upper half of the sixth thoracic vertebra, approximately 3 cm inferior to the sternal angle. (Redrawn with permission from Mirjalili S, Hale S, Buckenham T, Wilson B, Stringer M. 2012. A reappraisal of adult thoracic surface anatomy. Clin Anat 25:827–34.)

Table 51.1 The level of key thoracic viscera relative to the sternal plane and vertebrae

	Mean position relative to sternal plane (Range, cm)	Most common position/level (frequency % of 153 subjects studied)	Sex differences (95% CI)
Tracheal bifurcation	–3.1 cm (0 to –7.5)	T6 (46%) (lower T4 to lower T7)	1.3 cm higher in females (0.8–1.8 cm: $p < 0.001$)
Aortic arch concavity	–1.1 cm (–2.8 to –4.8)	T5 (55%) (lower T3 to lower T6)	0.6 cm higher in females (0.1–1.1 cm: $p = 0.01$)
Azygos vein joining the superior vena cava	–2 cm (+1.3 to –6.4)	T5 (49%) (upper T4 to upper T7)	0.7 cm higher in females (0.3–1.2 cm: $p = 0.001$)
Pulmonary trunk bifurcation	Not reported	T6 upper (28%) (T4/5 to T7/8)	No significant effect

Measurements relative to the sternal plane are shown as + (superior to the plane) and – (inferior to the plane). Adapted from Mirjalili S, Hale S, Buckenham T, Wilson B, Stringer M. 2012. A reappraisal of adult thoracic surface anatomy. Clin Anat. Oct;25(7):827–34.

latter point, as represented by the junction between the superior vena cava and right atrium, most commonly between the fourth and fifth right costal cartilages. The right border is a gently curved line, convex to the right, running from the third to the sixth right costal cartilages, usually 1–2 cm lateral to the sternal edge. The inferior or acute border runs leftwards from the sixth right costal cartilage to the cardiac apex, located approximately 9 cm lateral to the midline, often in the left fifth intercostal space or level with the fifth or sixth rib. Clinicians vary when locating the mid-clavicular line (Naylor et al 1987); a measurement from the midline is preferable. The most inferolateral point at which a pulsation is visible and palpable is called the cardiac apex beat and is usually palpable near the cardiac apex.

The left ('obtuse') border is convex laterally and extends superomedially from the cardiac apex to meet the second left costal cartilage approximately 1 cm from the left sternal edge. An oblique line joining the sternal end of the third left and sixth right costal cartilages

represents the anterior part of the coronary sulcus/atrioventricular groove, which separates the right atrium from the right ventricle. The projections of the cardiac valves, although in different planes, are also sited along or close to this line (see Fig. 57.19). The orifice of the pulmonary valve is represented by a horizontal line approximately 2.5 cm long that lies over the superior border of the left third costal cartilage and behind the corresponding region of the sternum. The pulmonary trunk is delineated by two parallel lines drawn perpendicular to the ends of the pulmonary valve line, up to the level of the left second intercostal space. The orifice of the aortic valve is located below and to the right of the orifice of the pulmonary valve, and is represented by a line approximately 2.5 cm long running inferolaterally and to the right from the medial end of the left third intercostal space. Two parallel lines, drawn perpendicular to the ends of the valve line and slanting up towards the right half of the manubriosternal joint, outline the location of the ascending aorta.

The orifice of the tricuspid valve is represented by a line approximately 4 cm long that starts near the midline just below the level of the right fourth costal cartilage and passes down and slightly to the right. The centre of this line should be level with the middle of the right fourth intercostal space. The orifice of the mitral valve is level with the left half of the sternum opposite the left fourth costal cartilage and is represented by a line approximately 3 cm long and descending to the right.

The direction of blood flow through a valve carries the valve sound with it and thus the sites of cardiac valve auscultation do not correspond to the surface anatomy of the valves. Convenient sites at which to auscultate using a stethoscope are: the sternal end of the left second intercostal space (pulmonary area); the sternal end of the right second intercostal space (aortic area); near the cardiac apex (mitral area); and over the left lower sternal border, at the level of the fifth intercostal space (tricuspid area) (see Fig. 57.19).

Great vessels

The aortic arch lies mainly posterior to the manubrium (see Fig. 51.6). The concavity of the arch sits approximately 1 cm inferior to the sternal plane, level with the upper half of the fifth thoracic vertebra (range 2.8 cm superior to 4.8 cm inferior). Starting at the aortic valve (see above), the ascending aorta curves anteriorly, superiorly and to the right, and becomes the aortic arch posterior to the right half of the manubrium at the level of the right second costal cartilage. It continues to ascend to the right side of the manubrium sterni, then arches to the left across or over the sternal plane and descends such that the aortic knuckle protrudes just to the left of the manubrium sterni around the first intercostal space. The brachiocephalic trunk arises approximately posterior to the centre point of the manubrium sterni and ascends to the right sternoclavicular joint. The left and right brachiocephalic veins are formed posterior to the sternoclavicular joints; the right brachiocephalic vein descends almost vertically, whereas the left brachiocephalic vein passes posterior to the superior portion of the manubrium sterni. According to conventional descriptions, the superior vena cava forms posterior to the right first costal cartilage and descends vertically to enter the right atrium at the level of the right third costal cartilage. CT data show that it commonly forms posterior to the right second costal cartilage or first intercostal space and the associated part of the manubrium sterni, and joins the right atrium between the right fourth and fifth costal cartilages; the level at which it forms is higher in females and younger adults. The azygos vein enters the superior vena cava approximately 2 cm inferior to the sternal plane at the level of the lower part of the fifth thoracic vertebra.

Lungs and pleurae

Anteriorly, the apex of the lung sits posterior to the medial third of the clavicle; posteriorly, it lies level with the seventh cervical vertebra (see Fig. 51.4). The surface markings of both the apex and costovertebral border of the lung correspond to those of the parietal pleura. Conventionally, the surface projection of the lower border of the lung is represented by a curved line that crosses the mid-clavicular line at the sixth rib, the mid-axillary line at the eighth rib and the tenth rib just lateral to the vertebral column. CT data reveal considerable variation, redefining the lower lung border, at the end of tidal inspiration, as lying mainly behind the fifth rib/fifth intercostal space on the left, or the sixth intercostal space/seventh rib on the right in the mid-clavicular line, and adjacent to the twelfth thoracic vertebra posteriorly (ranging from the ninth thoracic to the first lumbar vertebra). In full expiration, the lower margin of the lung may rise 5 cm above the parietal pleural

reflection (the costodiaphragmatic recess) and may retreat from the retrosternal costomediastinal recess.

Trachea and bronchi

The trachea starts at the level of the inferior border of the cricoid cartilage, which usually sits at the level of the seventh cervical vertebra (range C5 to T1) (Mirjalili et al 2012b). It runs inferiorly in the midline and bifurcates level with the upper half of the sixth thoracic vertebra, i.e. inferior to the sternal angle, and lower than was previously believed (see above). The trachea is palpable immediately superior to the suprasternal notch either by rolling one finger over its convex surface or by defining its lateral extents with two fingers. A shift to one side indicates that the mediastinum has either been pulled over to that side by loss of lung volume, or pushed over by a pneumothorax or pleural effusion. The level at which the trachea bifurcates moves down, on average, by a mean of 3 mm per decade.

Pulmonary fissures

On either side, the upper and lower lobes of the lung are separated by the oblique fissure. This can be marked by a line that runs anteroinferiorly from the posterior end of the fourth rib (spinous process of the third thoracic vertebra), crosses the fifth rib in the mid-axillary line and continues inferiorly, crossing either the sixth intercostal space/seventh rib on the right or the fifth rib/intercostal space on the left, on both sides at the mid-clavicular line (i.e. 7–8 cm lateral to the midline). As a convenient approximation, the oblique fissure follows the medial border of the scapula when the upper limb is in full abduction. The left oblique fissure may be slightly more vertical than the right.

The horizontal fissure lies between the upper and middle lobes of the right lung (see Fig. 51.4; Fig. 51.7). It extends from the fourth costal cartilage at the right sternal border to intersect the oblique fissure. The right middle lobe may be projected on to the thoracic wall using three points, namely: the fourth costal cartilage at the right parasternal edge, the fifth rib at the mid-axillary line and the sixth intercostal space/seventh rib at the mid-clavicular line. The upper and middle lobes lie anterior to the oblique fissure and are best examined from in front and to the side. The lower lobes lie posteriorly and should be examined from behind.

Pleural reflections

Starting in the midline at the sternal angle, the anterior reflections of the parietal pleura may be traced superiorly along a curved line that diverges from the midline and extends up and outwards to the apex of the lung and pleural cavity (see Fig. 51.4). The line lies 3–4 cm above the anterior end of the first rib; the surface marking is posterior to the

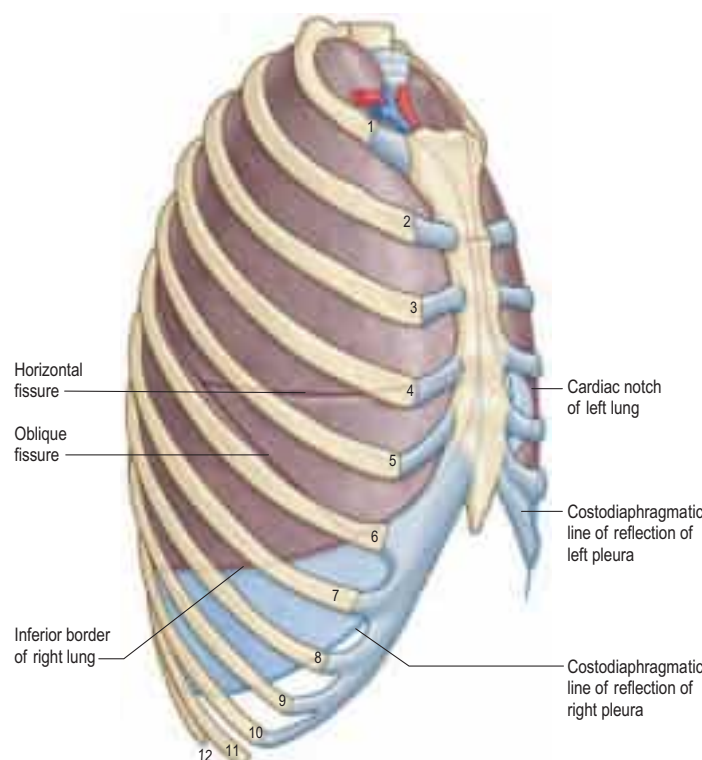


Fig. 51.7 The relation of the pleura and lungs to the chest wall, right lateral aspect.

medial third of the clavicle. The right and left pleurae are in contact retrosternally in the midline from the second to the fourth costal cartilages, at which point they diverge. The pleura on the right descends vertically to the xiphisternum, whereas the pleura on the left deviates laterally by 3–5 cm and then passes inferiorly to cross the anterior end of the sixth rib. This deviation produces an area between the heart and the sternum that is free of pleura; a needle puncture of the heart may be performed at this site without risk of damaging the pleura.

The costodiaphragmatic reflections of the pleurae may be followed around the chest wall from the sixth rib/costal cartilage anteriorly to the eighth rib in the mid-clavicular line, tenth rib in the mid-axillary line and twelfth rib, or just below, adjacent to the vertebral column. Viewed from behind, the medial edges of the pleurae may be followed along a line joining the transverse processes of the second to the twelfth thoracic vertebrae on either side.

The parietal pleura is fused with the inner aspect of the thoracic cavity and can be followed laterally and inferiorly down the inner aspect of the thoracic wall to the level of the tenth rib in the mid-axillary line

(see Fig. 51.4). The position of the diaphragmatic pleura varies according to the phase of ventilation.

Diaphragm

The dome of the diaphragm is often located level with the fifth intercostal space on the right and the sixth rib on the left; it ranges widely, from the fourth intercostal space to below the costal margin. Conventionally, the diaphragm is said to be traversed by the inferior vena cava at about the level of the disc between the eighth and ninth thoracic vertebrae, the oesophagus at the level of the tenth thoracic vertebra and the aorta at the level of the lower border of the twelfth thoracic vertebra and the adjacent disc, slightly to the left of the midline. CT assessment reveals that the inferior vena cava and the oesophagus frequently pass through the diaphragm at levels slightly lower than those predicted from chest radiographs, at the level of the eleventh thoracic vertebra (Mirjalili et al 2012a).

KEY REFERENCES

Hale SJ, Mirjalili SA, Stringer MD 2010 Inconsistencies in surface anatomy: the need for an evidence-based reappraisal. *Clin Anat* 23:922–93. *Analyses the consistency of modern clinical and surface anatomy texts in their reporting of common surface anatomy landmarks and calls for an evidence-based reappraisal of clinically relevant surface projections that takes account of the influence of age, sex, body mass, posture, respiration and ethnicity.*

Mirjalili S, Hale S, Buckenham T et al 2012a A reappraisal of adult thoracic surface anatomy. *Clin Anat* 25:827–34. *Examines key thoracic surface anatomical landmarks in vivo using computed tomographic (CT) imaging.*

REFERENCES

- Beckenstein MS, Windle BH, Stroup RT 1996 Anatomical parameters for nipple position and areolar diameter in males. *Ann Plast Surg* 36: 33–6.
- Boussuges A, Gole Y, Blanc P 2009 Diaphragmatic motion studied by M-mode ultrasonography: methods, reproducibility, and normal values. *Chest* 135:391–400.
- Hale SJ, Mirjalili SA, Stringer MD 2010 Inconsistencies in surface anatomy: the need for an evidence-based reappraisal. *Clin Anat* 23:922–93. *Analyses the consistency of modern clinical and surface anatomy texts in their reporting of common surface anatomy landmarks and calls for an evidence-based reappraisal of clinically relevant surface projections that takes account of the influence of age, sex, body mass, posture, respiration and ethnicity.*
- Macklin CC 1925 X-ray studies on bronchial movements. *Am J Anat* 35: 303–29.
- Mirjalili S, Hale S, Buckenham T et al 2012a A reappraisal of adult thoracic surface anatomy. *Clin Anat* 25:827–34. *Examines key thoracic surface anatomical landmarks in vivo using computed tomographic (CT) imaging.*
- Mirjalili SA, McFadden SL, Buckenham T et al 2012b Vertebral levels of key landmarks in the neck. *Clin Anat* 25:851–7.
- Naylor C, McCormack D, Sullivan S 1987 The midclavicular line: a wandering landmark. *Can Med Assoc J* 136:48–50.
- Takazakura R, Takahashi M, Nitta N et al 2004 Diaphragmatic motion in the sitting and supine positions: healthy subject study using a vertically open magnetic resonance system. *J Magn Reson Imaging* 19:605–60.